SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

---

COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning, Data intergration,Data transformation and data Reduction , for effective data preparation (BL3,BL4)
Assessment Tool Weightage: 50%

QUESTIONS: 5 **Total Marks:50**

Annexure B
Analytical Problem Solving

Criteria	Problem Understanding ***(2)***	Method Selection ***(2)***	Solution Process ***(2.5)***	Accuracy of Calculation ***(2)***	Interpretation of Results ***(1.5)***	***Total(10)***
Weightage	*20%*	20%	25%	20%	15%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I Hemalakshmi H (192571049) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Numerical Problems					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Annexure B
Analytical Problem Solving

1. Covariance Analysis in Health Insurance (10 Marks)

The table below shows the **annual premium increase rate (xi)** and the **number of claims per 100 policyholders (yi)** in a health insurance company.

Using the covariance formula, determine whether premium increase and claim frequency have a **positive or inverse relationship**.
Before computing covariance, calculate the **mean of x and y**.

Premium Increase % (xi)	Claims per 100 Policyholders (yi)
3.2	15
4.0	18
5.5	25
4.8	20

---

2. Data Smoothing using Equal-Frequency Binning (Healthcare) (5 Marks)

A hospital is analyzing **patient blood pressure readings (systolic)** to identify trends. Due to minor fluctuations, the analyst performs **local smoothing using equal-frequency (equi-depth) binning**.

Sample Data Set (Systolic BP readings):
110, 112, 115, 115, 118, 120, 120, 122, 124, 124, 126, 126, 126, 126, 130, 135, 135, 138, 138, 138, 138, 140, 145, 150, 152, 160, 180

---

3. Equal-Width Binning for Risk Categorization (5 Marks)

An insurance company evaluates **Body Mass Index (BMI)** of 18 clients for risk classification.
To simplify analysis, they smooth the data by replacing values with **closest bin boundaries using equal-width partitioning**.

Sample Data Set (Sorted BMI):
18.5, 19.2, 21.8, 24.9, 25.5, 26.2, 26.4, 27.8, 29.2, 30.2, 30.4, 31.9, 32.4, 33.1, 33.6, 34.7, 38.2, 39.5

---

4. Data Reduction using Bin Medians (Medical Supply Costs) (5 Marks)

A hospital administration is reviewing the **cost of medical consumables (in ₹)**.
To reduce data for mining purposes, they discretize the sorted price records using **bin medians**.

Sample Data Set (Costs in ₹):
120, 150, 160, 160, 140, 145, 145, 148, 150, 150, 155, 155, 155, 155, 165, 170, 170, 175, 175, 175, 175, 180, 190, 200, 210, 230, 300

---

5. Standardization of Patient Age Data (5 Marks)

Given the following **ages of insured individuals**:

18, 22, 25, 42, 28, 43, 33, 35, 56, 28

Standardize the variable by performing the following:

(a) Compute the **mean absolute deviation (MAD)** of age.
(b) Compute the **z-score for the first four observations**.

Q1: Answer

Step 1: Write down the data

- Premium increase x_i: 3.2, 4.0, 5.5, 4.8
- Claims per 100 policyholders y_i: 15, 18, 25, 20

Number of observations: $n = 4$

Step 2: Calculate the mean of x

$$\bar{x} = \frac{\sum x_i}{n} = \frac{3.2 + 4.0 + 5.5 + 4.8}{4}$$

$$3.2 + 4.0 = 7.2$$

$$7.2 + 5.5 = 12.7$$

$$12.7 + 4.8 = 17.5$$

$$\bar{x} = \frac{17.5}{4} = 4.375$$

So, Mean premium increase $\bar{x} = 4.375\%$.

Step 3: Calculate the mean of y

$$\bar{y} = \frac{\sum y_i}{n} = \frac{15 + 18 + 25 + 20}{4}$$

$$15 + 18 = 33$$

$$33 + 25 = 58$$

$$58 + 20 = 78$$

$$\bar{y} = \frac{78}{4} = 19.5$$

So, Mean claims $\bar{y} = 19.5$per 100 policyholders.

Step 4: Find deviations from the mean

For each pair (x_i, y_i), compute:

$$x_i - \bar{x},\ y_i - \bar{y}$$

1. First pair $(3.2,\ 15)$

$$x_1 - \bar{x} = 3.2 - 4.375 = -1.175$$

$$y_1 - \bar{y} = 15 - 19.5 = -4.5$$

2. Second pair $(4.0,\ 18)$

$$x_2 - \bar{x} = 4.0 - 4.375 = -0.375$$

$$y_2 - \bar{y} = 18 - 19.5 = -1.5$$

3. Third pair $(5.5,\ 25)$

$$x_3 - \bar{x} = 5.5 - 4.375 = 1.125$$

$$y_3 - \bar{y} = 25 - 19.5 = 5.5$$

4. Fourth pair $(4.8,\ 20)$

$$x_4 - \bar{x} = 4.8 - 4.375 = 0.425$$

$$y_4 - \bar{y} = 20 - 19.5 = 0.5$$

Step 5: Multiply deviations for each pair

Covariance uses:

$$\text{Cov}(x, y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n}$$

Compute each product:

1. First pair

$$(x_1 - \bar{x})(y_1 - \bar{y}) = (-1.175)(-4.5) = 5.2875$$

2. Second pair

$$(x_2 - \bar{x})(y_2 - \bar{y}) = (-0.375)(-1.5) = 0.5625$$

3. Third pair

$$(x_3 - \bar{x})(y_3 - \bar{y}) = (1.125)(5.5) = 6.1875$$

4. Fourth pair

$$(x_4 - \bar{x})(y_4 - \bar{y}) = (0.425)(0.5) = 0.2125$$

Now sum them:

$$\sum (x_i - \bar{x})(y_i - \bar{y}) = 5.2875 + 0.5625 + 6.1875 + 0.2125 = 12.25$$

Step 6: Compute covariance

Using the population covariance formula:

$$\text{Cov}(x, y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n} = \frac{12.25}{4} = 3.0625$$

So, Covariance $\text{Cov}(x, y) = 3.0625$.

Step 7: Interpret the relationship

- If covariance > 0 → positive relationship
- If covariance < 0 → inverse (negative) relationship
- Here: $\text{Cov}(x, y) = 3.0625 > 0$

Conclusion: Premium increase and claim frequency have a positive relationship—as premium increase percentage rises, the number of claims per 100 policyholders also tends to increase.

Q2: Answer

Step 1: Sorted Data

110, 112, 115, 115, 118, 120, 120, 122, 124, 124, 126, 126, 126, 126, 130, 135, 135, 138, 138, 138, 138, 140, 145, 150, 152, 160, 180
Total values = 27

Step 2: Number of Bins

Equal-frequency binning with 3 bins.
Values per bin = 27 ÷ 3 = 9

Step 3: Create the Bins

Bin 1 (first 9 values)
110, 112, 115, 115, 118, 120, 120, 122, 124 Boundaries: 110 to 124
Bin 2 (next 9 values)
124, 126, 126, 126, 126, 130, 135, 135, 138 Boundaries: 124 to 138
Bin 3 (last 9 values)
138, 138, 138, 140, 145, 150, 152, 160, 180 Boundaries: 138 to 180

Step 4: Compute Bin Means

Bin 1 Mean
(110 + 112 + 115 + 115 + 118 + 120 + 120 + 122 + 124) ÷ 9 = 1056 ÷ 9 = 117.33
Bin 2 Mean
(124 + 126 + 126 + 126 + 126 + 130 + 135 + 135 + 138) ÷ 9 = 1176 ÷ 9 = 130.67
Bin 3 Mean
(138 + 138 + 138 + 140 + 145 + 150 + 152 + 160 + 180) ÷ 9 = 1341 ÷ 9 = 149.00

Step 5: Smoothed Values Using Bin Means

Bin 1 Smoothed

117.33 repeated 9 times

Bin 2 Smoothed

130.67 repeated 9 times

Bin 3 Smoothed

149.00 repeated 9 times

Q3: Answer

Step 1: Write down the data and basic info

Sorted BMI values (18 clients): 18.5, 19.2, 21.8, 24.9, 25.5, 26.2, 26.4, 27.8, 29.2, 30.2, 30.4, 31.9, 32.4, 33.1, 33.6, 34.7, 38.2, 39.5

- **Minimum (min)** = 18.5
- **Maximum (max)** = 39.5
- **Range** = max − min = 39.5 − 18.5 = 21.0

Assume **3 equal-width bins** (common choice in such questions).

Step 2: Compute bin width

$$\text{Bin width} = \frac{\text{Range}}{\text{Number of bins}} = \frac{21.0}{3} = 7.0$$

Step 3: Define bin boundaries

Starting from min = 18.5, width = 7.0:

- **Bin 1:** 18.5 to 25.5
- **Bin 2:** 25.5 to 32.5
- **Bin 3:** 32.5 to 39.5

Step 4: Assign values to bins

Bin 1 (18.5–25.5): 18.5, 19.2, 21.8, 24.9, 25.5

Bin 2 (25.5–32.5): 26.2, 26.4, 27.8, 29.2, 30.2, 30.4, 31.9, 32.4

Bin 3 (32.5–39.5): 33.1, 33.6, 34.7, 38.2, 39.5

Step 5: Smoothing by closest bin boundaries

We replace each value with the **closest boundary of its bin**.

Bin 1 boundaries: 18.5 and 25.5

- 18.5 → 18.5 (already boundary)
- 19.2 → closer to 18.5 (|19.2−18.5|=0.7, |19.2−25.5|=6.3) → 18.5
- 21.8 → closer to 18.5 (3.3 vs 3.7) → 18.5
- 24.9 → closer to 25.5 (0.6 vs 6.4) → 25.5
- 25.5 → 25.5 (boundary)

Smoothed Bin 1: 18.5, 18.5, 18.5, 25.5, 25.5

Bin 2 boundaries: 25.5 and 32.5

- 26.2 → closer to 25.5 (0.7 vs 6.3) → 25.5
- 26.4 → closer to 25.5 (0.9 vs 6.1) → 25.5
- 27.8 → closer to 25.5 (2.3 vs 4.7) → 25.5
- 29.2 → closer to 32.5 (3.7 vs 3.3) → 32.5
- 30.2 → closer to 32.5 (4.7 vs 2.3) → 32.5
- 30.4 → closer to 32.5 (4.9 vs 2.1) → 32.5
- 31.9 → closer to 32.5 (6.4 vs 0.6) → 32.5
- 32.4 → closer to 32.5 (6.9 vs 0.1) → 32.5

Smoothed Bin 2: 25.5, 25.5, 25.5, 32.5, 32.5, 32.5, 32.5, 32.5

Bin 3 boundaries: 32.5 and 39.5

- 33.1 → closer to 32.5 (0.6 vs 6.4) → 32.5
- 33.6 → closer to 32.5 (1.1 vs 5.9) → 32.5
- 34.7 → closer to 32.5 (2.2 vs 4.8) → 32.5
- 38.2 → closer to 39.5 (5.7 vs 1.3) → 39.5
- 39.5 → 39.5 (boundary)

Smoothed Bin 3: 32.5, 32.5, 32.5, 39.5, 39.5

Step 6: Final smoothed BMI values (in original order)

Original: 18.5, 19.2, 21.8, 24.9, 25.5, 26.2, 26.4, 27.8, 29.2, 30.2, 30.4, 31.9, 32.4, 33.1, 33.6, 34.7, 38.2, 39.5

Smoothed (closest bin boundaries): 18.5, 18.5, 18.5, 25.5, 25.5, 25.5, 25.5, 25.5, 32.5, 32.5, 32.5, 32.5, 32.5, 32.5, 32.5, 32.5, 32.5, 39.5, 39.5

This is the equal-width binning with boundary-based smoothing, suitable for risk categorization.

Q4: Answer:

Step 1: Sort the Data

Given data (unsorted): 120, 150, 160, 160, 140, 145, 145, 148, 150, 150, 155, 155, 155, 155, 165, 170, 170, 175, 175, 175, 175, 180, 190, 200, 210, 230, 300
Sorted data:
120, 140, 145, 145, 148, 150, 150, 150, 155, 155, 155, 155, 160, 160, 165, 170, 170, 175, 175, 175, 175, 175, 180, 190, 200, 210, 230, 300
Total values = **27**

Step 2: Decide Number of Bins

Common choice: **3 bins**
Values per bin:

$$\frac{27}{3} = 9$$

So each bin contains **9 values**.

Step 3: Create the Bins (Equal Frequency)

Bin 1 (first 9 values)
120, 140, 145, 145, 148, 150, 150, 150, 155 Median = 148
Bin 2 (next 9 values)
155, 155, 155, 160, 160, 165, 170, 170, 175 Median = 160
Bin 3 (last 9 values)
175, 175, 175, 180, 190, 200, 210, 230, 300 Median = 190

Step 4: Replace Each Value with Bin Median

Bin 1 smoothed (median = 148)

148, 148, 148, 148, 148, 148, 148, 148, 148

Bin 2 smoothed (median = 160)

160, 160, 160, 160, 160, 160, 160, 160, 160

Bin 3 smoothed (median = 190)

190, 190, 190, 190, 190, 190, 190, 190, 190

Final Smoothed Dataset (in original order)

148, 148, 148, 148, 148, 148, 148, 148, 148, 160, 160, 160, 160, 160, 160, 160, 160, 190, 190, 190, 190, 190, 190, 190, 190, 190

Conclusion

Using **bin medians**, the hospital reduces the dataset into three stable representative cost levels:

- Low-cost consumables ≈ **₹148**
- Mid-range consumables ≈ **₹160**
- High-cost consumables ≈ **₹190**

This reduces noise and simplifies further mining or trend analysis.

If you want, I can also prepare **bin means**, **bin boundaries**, or **equal-width bins** for the same dataset.

Q5: Answer

Given Ages

18, 22, 25, 42, 28, 43, 33, 35, 56, 28 Number of observations = 10

(a) Mean Absolute Deviation (MAD)

Step 1: Compute the Mean

Sum of ages:

18 + 22 = 40 40 + 25 = 65 65 + 42 = 107 107 + 28 = 135 135 + 43 = 178 178 + 33 = 211 211 + 35 = 246 246 + 56 = 302 302 + 28 = 330

Mean:

$$\bar{x} = \frac{330}{10} = 33$$

Step 2: Compute Absolute Deviations from Mean

Sum of absolute deviations:

15 + 11 + 8 + 9 + 5 + 10 + 0 + 2 + 23 + 5 = **88**

MAD

$$\text{MAD} = \frac{88}{10} = 8.8$$

Mean Absolute Deviation = 8.8

(b) Compute Z-Scores for First Four Observations

Z-score formula:

$$z = \frac{x - \bar{x}}{s}$$

We need the **standard deviation**.

Step 1: Compute Squared Deviations

Age	**(Age − Mean)**	**(Age − Mean)²**
18	−15	225
22	−11	121
25	−8	64
42	9	81
28	−5	25
43	10	100
33	0	0
35	2	4
56	23	529
28	−5	25

Sum of squared deviations:

225 + 121 + 64 + 81 + 25 + 100 + 0 + 4 + 529 + 25 = **1174**

Population Variance

$$\sigma^2 = \frac{1174}{10} = 117.4$$

Standard Deviation

$$\sigma = \sqrt{117.4} \approx 10.834$$

Step 2: Z-Scores for First Four Ages

1. Age = 18

$$z = \frac{18 - 33}{10.834} = \frac{-15}{10.834} \approx -1.384$$

2. Age = 22

$$z = \frac{22 - 33}{10.834} = \frac{-11}{10.834} \approx -1.015$$

3. Age = 25

$$z = \frac{25 - 33}{10.834} = \frac{-8}{10.834} \approx -0.739$$

4. Age = 42

$$z = \frac{42 - 33}{10.834} = \frac{9}{10.834} \approx 0.831$$

Final Answers

(a) Mean Absolute Deviation (MAD): 8.8

(b) Z-scores for first four ages:

- 18 → **−1.384**
- 22 → **−1.015**
- 25 → **−0.739**
- 42 → **0.831**